

Overview of Vertical Issues

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Grad IO

Overview

- ▶ If you haven't worked through double marginalization, read Chapter 4 of Tirole
- ▶ You should read the chapter from Whinston's antitrust lectures.
- ▶ Exclusionary Contracts: Conlon Mortimer (JPE 2021)
- ▶ Bargaining Models in Healthcare: Ho and Lee (AER 2019), Gowrisankaran, Nevo Town (AER 2015).
- ▶ Bargaining in TV: Crawford, Lee, Whinston, Yurukoglou (ECMA 2018), Crawford Yurukoglou (AER 2012)
- ▶ Do upstream or downstream firms set prices (Villas Boas (2007), Bonnet and Dubois (2010)).

Manufacturers rarely supply final consumers directly (as we have modeled them so far). Instead, most industries are vertically separated.

We often refer to firms in these markets as upstream and downstream firms. In these settings, downstream firms are the customers of the upstream firms, and many of the standard issues still apply. For example:

1. choice of price is endogenous
2. price discrimination (both the upstream and downstream firms)
3. mergers
4. entry, etc.

However, things can also get more complicated in vertically separated environments. In particular, downstream firms do not usually consume the good, but typically make further decisions regarding the product.

Examples of activities of downstream firms:

1. determination of final price
2. promotional effort
3. placement of product on store shelves
4. promotion and placement of competing products
5. technological inputs

Why don't manufacturers simply engage in direct marketing to consumers?

Some reasons:

1. increasing returns to distribution due to shopping needs or travel costs for consumers
2. choice of variety
3. demand for service
4. integration of complementary products
5. different geographical markets, etc.

Vertical Control

Unlike the consumption activities of final consumers, the activities of the downstream firms may affect the profits of the upstream firm.

This is why upstream firms care about the activities of the downstream firms, and why we study vertical control/restraints between firms in these settings.

We focus on the incentives for vertical control when the market for the intermediate good is imperfectly competitive.

A common benchmark for what firms can achieve through vertical control is the “vertically integrated profit.” This is the maximum industry or aggregate (manufacturer plus retailer) profit.

If firms use vertical restraints efficiently, they should achieve the vertically integrated profit.

There are several types of vertical restraints used by firms in vertically-separated markets:

1. *Exclusive Territories*: a dealer/ distributor/ retailer is assigned a (usually geographic) territory by the manufacturer/ upstream firm and given monopoly rights to sell in that area. [e.g Car Dealerships, Franchises, Beverage Distributors]
2. *Exclusive Dealing*: a dealer/ distributor/ retailer is not allowed to carry the brands of a competing upstream firm.[See Asker 2005 or Sass 2004,2005] on beer distribution.
3. *Full-line forcing*: a dealer is committed to sell all the varieties of the manufacturer's products rather than a limited selection. (i.e., the upstream firm ties all its products to sell to the downstream firm). See [Ho, Ho, Mortimer AER 2012] on video rentals.

4. *Resale Price Maintenance*: a dealer commits to a retail price or a range of retail prices for the product. This can take the form of either minimum resale price maintenance or maximum resale price maintenance. Equivalently, firms can engage in quantity forcing or quantity rationing. [Apple (appears to have) minimum RPM for its products]
5. *Contractual arrangements*: upstream and downstream firms write contracts to provide greater flexibility in the transfer of the product. Profit sharing and revenue sharing [Mortimer ReStud 2008] are the most common, which we'll see soon. Also, franchising arrangements.

- ▶ There are many ambiguities in the legal treatment of vertical contracts.
- ▶ Until 1970s, RPM and E. Territories were per se illegal under Sherman Act.
- ▶ But many states passed fair trade laws that were interpreted to cover some of these cases.
- ▶ Furthermore, the Khan case in 1997 switched Maximum RPM to a “rule of reason” status, as did the Leegin Leather Products case in 2007 for Minimum RPM.

Thus, although price fixing remains per se illegal, it's not always applied in vertical settings because it conflicts with free-trade notions between mfgs and their distributors.

Non-price issues have been generally accepted to be ok by the courts. Decisions turn on arguments about efficiency vs. anti-competitive effects.

- ▶ Exclusive territories
- ▶ Refusal to deal
- ▶ Foreclosure, etc.

Further Reading

Understanding the legal framework is important for working with vertical restraints:

- ▶ John Kwoka and Larry White (NYU Stern) have compiled an edited volume of summarized IO economist testimony and reports from various cases in *The Antitrust Revolution* across soon to be 7 volumes.
- ▶ Whinston has written some nice lectures on Antitrust, and Chapter 4 focuses specifically on vertical issues.
- ▶ There is not much on legal details, but Tirole's *Theory of Industrial Organization* covers many theoretical models on vertical contracting.

The typical outline of vertical control is as follows:

1. Double Marginalization/Successive Monopoly Problem.
2. Externalities between downstream and upstream firms (Maximum Resale Price Maintenance, Quantity Forcing, Contractual Arrangements, or Full-line Forcing)
3. Downstream Moral Hazard, or Externalities from Intrabrand competition (Exclusive Territories, Minimum RPM, or Quantity Rationing)
4. Interbrand competition (Exclusive Dealing or possibly Full-line Forcing)

- ▶ The DOJ is still operating under the June 2020 Vertical Merger Guidelines
- ▶ The FTC has rescinded the guidelines
- ▶ Controversy:
 - The 2020 guidelines (read them) stressed **elimination of double marginalization** (EDM) in vertical mergers as an **efficiency** argument.
 - Perhaps not enough discussion of potential foreclosure/raising rivals costs (More on that later).

Basic Framework

Double Marginalization

- ▶ U sells product to D who sells product to final consumers.
- ▶ Homogeneous product with final demand given by $D(p_d)$.
- ▶ U charges p_w to D .
- ▶ D chooses how much to buy from U

$$\max_{p_d} \pi_d \equiv \max_{p_d} (p_d - p_w) D(p_d)$$

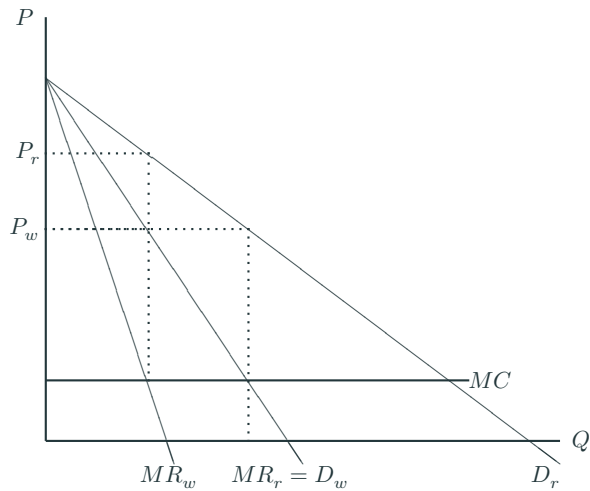
$$\text{FOC:} \quad (p_d - p_w) D'(p_d) + D(p_d) = 0$$

- ▶ The solution is denoted by $p_d^*(p_w)$
- ▶ The upstream firm (U) now solves:

$$\max_{p_w} \pi_w \equiv \max_{p_w} (p_w - c) D(p_d^*(p_w))$$

- ▶ We have that $p_w > c$ and $p_d > p_w$ which we call **double marginalization**.

Double Marginalization: In Pictures



- ▶ Double Marginalization arises from the externality as p_w raises his price this raises the effective marginal cost to U and the monopoly price she charges is too large.
- ▶ Think about a vertically integrated firm.

$$\begin{aligned}\max_{p_m} \pi_d &\equiv \max_{p_m} (p_m - c)D(p_m) \\ \text{FOC:} &\quad (p_m - c)D'(p_m) + D(p_m) = 0\end{aligned}$$

- ▶ Since $p_w > c$ then we have that $\pi_D + \pi_U < \pi_{VI}$ and $p_D^*(p_w) > p^*(c)$

Double Marginalization

Most solutions involve contracting around the externality:

- ▶ Consider a **two-part tariff** where $p_w = c$ but that $\widetilde{\pi}_D = \pi_D - T$ and $\widetilde{\pi}_U = \pi_U + T$. We call T the **franchise fee**. This is sometimes known as the **sell out contract** because the wholesaler sets $p_w = c$.
- ▶ Now we have that $\widetilde{\pi}_D = \pi_{VI} - T$ and $\widetilde{\pi}_U = 0 + T$.
- ▶ There are other (sometimes legal, sometimes not solutions): **RPM** to set $p_d^* = p^m$.
- ▶ **Quantity Forcing**. Upstream firm makes a TILO offer of monopoly quantity to D .
- ▶ We can also allow **revenue or profit sharing** where U “owns” a fraction λ of the upstream firm.
 - These contracts are common in franchises. (ie: Subway corporate keeps 20% of your revenue).

- ▶ Good empirical work on these topics is generally limited by the availability of data
- ▶ It is not too difficult to gather data on retail (P, Q) .
- ▶ Wholesale prices are harder to observe.
- ▶ Most nonlinear contracts between upstream and downstream firms are closely guarded trade secrets.
- ▶ If you can get your hands on contracts, you can write papers!

A long literature relates the pass-through matrix to the curvature of demand (2nd derivatives)

- ▶ Bulow Pfleiderer (JPE 1983)
- ▶ Fabinger Weyl (JPE 2013)
- ▶ Recent work by Eugenio Miravete and Katja Seim.
- ▶ But estimating PTR directly from data can be tough – assumes smooth transmission of cost shocks (no menu prices, etc.). See Conlon Rao AEJ:Pol (2019).

- ▶ There is recent work empirical work on **vertical restraints**.
- ▶ Conlon and Mortimer (JPE 2021)
 - This paper asks how an **upstream firm** can use contracts to **exclude** an upstream rival from selling his products via a downstream retailer.
 - The focus is on fully categorizing the set of exclusionary rebate contracts that: (a) the dominant firm is willing to offer (b) the retailer is willing to sign (thus excluding the rival) and (c) the rival is unwilling to deviate to prevent exclusion.
 - The authors address the potential welfare implications of such contracts in a world with **downstream moral hazard** so that exclusion may be efficient (or not).

- ▶ There also recent empirical work on **vertical integration**:
- ▶ Crawford, Lee, Whinston, Yurukoglou (ECMA 2018)
 - This paper asks how vertical integration changes the incentives for **downstream firms** to raise the price of upstream inputs to its downstream rivals.
 - The vertically integrated firm may **raise rivals costs** or it may fully **foreclose** its rival from acquiring the input.
 - Vertical integration may be good for efficiency reasons, but bad if foreclosure effects are large.
 - This approach builds on a literature using **Nash Bargaining** solutions to determine how to allocate surplus among upstream and downstream firms.